

Problem Set 5:

Abgabe: 12:00, Freitag, 15. Mai 2026; Lösungen: Freitag, 29. Mai 2026

Problem 1 Inverting Biot-Savart's Law

In the static limit and assuming the Coulomb gauge $\nabla \cdot \mathbf{A} = 0$, Maxwell's equation

$$\nabla \times \mathbf{B} - \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J}, \quad (1)$$

becomes the Poisson equation for the vector potential $\mathbf{A}$:

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}. \quad (2)$$

(1.a) Show that, based on this, the solution to this equation is given by the following expression:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3\mathbf{r}'. \quad (3)$$

Hint: You may assume the Green's function of Poisson's equation to be known.

(1.b) Use the result from part (a) to derive Biot-Savart's law:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} d^3\mathbf{r}'. \quad (4)$$

Let us now consider a specific system in which the magnetic field $\mathbf{B}(x, z)$ is generated by a surface current $\mathbf{K}(y, z) = K(z)\hat{\mathbf{y}}$ in the $x = x_0$ plane.

(1.c) Show that in this case, the Biot-Savart law reduces to a one-dimensional convolution integral of the function $K(z)$ with another function (to be determined by you).

(1.d) Now restrict yourself to the half-space $x < x_0$ and show that there

$$\mu_0 K(z) = \frac{1}{\pi} \int_{-\infty}^{\infty} dz' \int_{-\infty}^{\infty} dk \exp\{ik(z - z') + |k|(x_0 - x)\} B_z(x, z'). \quad (5)$$

holds.

Hint: Note the functional similarity to the expression you derived in problem 4 of the last problem set. Also recall the convolution theorem.

(1.e) Give an intuitive explanation of why the single component $B_z(x, z)$, evaluated at any point $x < x_0$, already contains enough information to determine $K(z)$.

Solution:

(1.a) We start from

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J},$$

and notice that this is just the Poisson equation component by component. The Green's function per component is

$$G(\mathbf{r}; \mathbf{r}') = \frac{1}{4\pi} \frac{1}{|\mathbf{r} - \mathbf{r}'|},$$

which gives the solution to $\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$ as the convolution with the source:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int d^3 \mathbf{r}' \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}.$$

(1.b) Note that the nabla in $\nabla \times \mathbf{A}$ only acts on $\mathbf{r}$ and not on $\mathbf{r}'$. We get

$$\begin{aligned} B_k &= (\nabla \times \mathbf{A})_k \\ &= \epsilon_{ijk} \frac{\mu_0}{4\pi} \int d^3 \mathbf{r}' J_j(\mathbf{r}') \partial_i \left(\frac{1}{|\mathbf{r} - \mathbf{r}'|} \right) \\ &= -\epsilon_{ijk} \frac{\mu_0}{4\pi} \int d^3 \mathbf{r}' J_j(\mathbf{r}') \frac{r_i - r'_i}{|\mathbf{r} - \mathbf{r}'|^3} \\ &= \frac{\mu_0}{4\pi} \int d^3 \mathbf{r}' \frac{[\mathbf{J}(\mathbf{r}') \times (\mathbf{r} - \mathbf{r}')]_k}{|\mathbf{r} - \mathbf{r}'|^3}. \end{aligned}$$

Hence,

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int d^3 \mathbf{r}' \frac{\mathbf{J}(\mathbf{r}') \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}.$$

(1.c) With

$$\mathbf{J}(\mathbf{r}') = \delta(x' - x_0) K(z') \hat{\mathbf{y}},$$

and simplifying,

$$\hat{\mathbf{y}} \times (\mathbf{r} - \mathbf{r}') = -(x - x')\hat{\mathbf{z}} + (z - z')\hat{\mathbf{x}}.$$

Then

$$\begin{aligned} \mathbf{B}(\mathbf{r}) &= \frac{\mu_0}{4\pi} \int d^3 \mathbf{r}' \delta(x' - x_0) K(z') \frac{-(x - x')\hat{\mathbf{z}} + (z - z')\hat{\mathbf{x}}}{|\mathbf{r} - \mathbf{r}'|^3} \\ &= \frac{\mu_0}{4\pi} \int dy' dz' K(z') \frac{-(x - x_0)\hat{\mathbf{z}} + (z - z')\hat{\mathbf{x}}}{((x - x_0)^2 + (y - y')^2 + (z - z')^2)^{3/2}} \\ &= \frac{\mu_0}{4\pi} \int dz' K(z') \int dy' \frac{-(x - x_0)\hat{\mathbf{z}} + (z - z')\hat{\mathbf{x}}}{((x - x_0)^2 + (y')^2 + (z - z')^2)^{3/2}}. \end{aligned}$$

We perform the integral over y' :

$$\int_{-\infty}^{+\infty} d\alpha \frac{1}{(a^2 + \alpha^2)^{3/2}} = \frac{2}{a^2},$$

to arrive at

$$\mathbf{B}(x, z) = \frac{\mu_0}{2\pi} \int_{-\infty}^{+\infty} dz' K(z') \frac{-(x - x_0)\hat{\mathbf{z}} + (z - z')\hat{\mathbf{x}}}{(x - x_0)^2 + (z - z')^2}.$$

This is indeed a convolution in z' .

- (1.d) Focus on $x < x_0$ and define $d = x_0 - x > 0$. We Fourier transform where the convolution theorem applies. Then

$$\widehat{B}_z(x, k) = \frac{\mu_0}{2} e^{-|k|d} \widehat{K}(k).$$

Here we used, just like we did in problem set 4, the Fourier transform of a Lorentzian. Therefore,

$$\mu_0 \widehat{K}(k) = 2e^{|k|d} \widehat{B}_z(x, k).$$

Transforming back,

$$\begin{aligned} \mu_0 K(z) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} dk e^{ikz} \left(2e^{|k|d} \widehat{B}_z(x, k) \right) \\ &= \frac{1}{\pi} \int_{-\infty}^{+\infty} dz' \int_{-\infty}^{+\infty} dk e^{ik(z-z') + |k|(x_0-x)} B_z(x, z'). \end{aligned}$$

- (1.e) The source $K(z)$ contains all the Fourier modes of the field according to

$$\mu_0 \widehat{K}(k) = 2e^{|k|d} \widehat{B}_z(x, k).$$

Given B_z on any line $x < x_0$, the solution is invertible, since $e^{|k|d} > 0$, and therefore B_z contains the same information as K .

Problem 2 (Zentralübung) The Aharonov–Bohm Effect

The Aharonov–Bohm effect shows that in quantum mechanics, not only magnetic and electric fields but also the vector potential can have physical significance. In particular, a charged particle can experience a *measurable* phase shift even though it moves only in regions where the magnetic field vanishes. What is crucial here is the magnetic flux through an inaccessible region, such as the interior of a long solenoid.

- (2.a) Determine the vector potential inside and outside a long solenoid that generates a magnetic field $\mathbf{B} = B\hat{\mathbf{z}}$ within an infinite cylinder of radius R . Work in the Coulomb gauge.
- (2.b) The Aharonov–Bohm effect occurs because the magnetic flux

$$\Phi_B = \oint d\mathbf{s} \cdot \mathbf{A} \quad (6)$$

does not vanish when the line of integration is, for example, along the edge of a disk of radius $\rho > R$ that lies perpendicular to the axis of the solenoid. Show that the gauge transformation

$$\mathbf{A}' = \mathbf{A} + \nabla\chi, \quad \chi = -\frac{\Phi\phi}{2\pi}, \quad (7)$$

where ϕ is the angle in cylindrical coordinates, leads to an identically vanishing vector potential outside the solenoid.

- (2.c) The result from part (b) suggests that the de Aharonov–Bohm effect could be eliminated by a gauge transformation. However, show that the new magnetic field (in the new gauge) corresponds to a different physical problem, namely

$$\mathbf{B}' = \mathbf{B} - \hat{\mathbf{z}} \frac{\Phi}{2\pi\rho} \delta(\rho). \quad (8)$$

Solution:

(2.a) We consider an infinitely long cylinder with radius R parallel to the z -axis. We seek a vector potential that generates the magnetic field

$$\mathbf{B} = B \hat{e}_z, \quad (9)$$

inside the cylinder.

Due to symmetry, we choose

$$\mathbf{A} = A_\varphi(s) \hat{\varphi}. \quad (10)$$

Then

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (11)$$

holds. For $s \leq R$

$$\frac{1}{s} \frac{\partial}{\partial s} (s A_\varphi(s)) \hat{e}_z = B \hat{e}_z \quad (12)$$

thus

$$\frac{\partial}{\partial s} (s A_\varphi(s)) = s B \quad (13)$$

Integration yields

$$s A_\varphi(s) = \int_0^s \frac{\partial}{\partial s'} (s' A_\varphi(s')) ds' = \int_0^s s' B ds' = \int s B ds \quad (14)$$

$$s A_\varphi(s) = \frac{1}{2} s^2 B, \quad (15)$$

therefore

$$A_\varphi(s) = \frac{1}{2} s B. \quad (16)$$

For $s \geq R$

$$\nabla \times \mathbf{A} = 0 \quad (17)$$

thus

$$\frac{\partial}{\partial s} (s A_\varphi(s)) = 0 \quad (18)$$

Integration gives:

$$s A_\varphi(s) = R A_\varphi(R) \quad (19)$$

Hence we obtain

$$\mathbf{A} = \begin{cases} \frac{1}{2} s B \hat{\varphi} & s \leq R \\ \frac{R^2 B}{2s} \hat{\varphi} & s > R \end{cases} \quad (20)$$

Furthermore,

$$\nabla \cdot \mathbf{A} = \nabla \cdot \hat{\varphi} = 0 \quad (21)$$

i.e. $\mathbf{A}$ is in the Coulomb gauge.

Alternative derivation:

The flux through a disk of radius r , D_r , is

$$\Phi_B(r) = \int_{D_r} \mathbf{B} \cdot d^2\mathbf{S} \quad (22)$$

$$= \int_0^r ds s \int_0^{2\pi} d\varphi \underbrace{\mathbf{B} \mathbf{e}_z}_{=B\Theta(R-s)} \quad (23)$$

$$= \begin{cases} \pi B r^2 & \text{if } r \leq R \\ \pi B R^2 & \text{if } r \geq R \end{cases} \quad (24)$$

At the same time,

$$\Phi_B(r) = \int_{D_r} \mathbf{B} \cdot d^2\mathbf{S} = \int_{D_r} \nabla \times \mathbf{A} \cdot d^2\mathbf{S} = \oint_{\partial D_r} \mathbf{A} \cdot d\mathbf{l} \quad (25)$$

On ∂D_r : $d\mathbf{l} = r d\varphi \mathbf{e}_\varphi$ and $\mathbf{A} = A_\varphi(r) \mathbf{e}_\varphi$. Thus

$$\Phi_B(r) = 2\pi r A_\varphi(r) \quad (26)$$

Comparing this with the previous expression for $\Phi_B(r)$ gives

$$A_\varphi(r) = \frac{\Phi_B(r)}{2\pi r} = \begin{cases} \frac{Br}{2} & \text{if } r \leq R \\ \frac{BR^2}{2r} & \text{if } r \geq R \end{cases} \quad (27)$$

(2.b) A gauge transformation has the form

$$\mathbf{A}' = \mathbf{A} + \nabla \chi \quad (28)$$

with

$$\chi(\varphi) = -\frac{\Phi_B}{2\pi} \varphi \quad (29)$$

In cylindrical coordinates,

$$\nabla \chi = \frac{1}{s} \frac{\partial \chi}{\partial \varphi} \hat{\varphi} \quad (30)$$

therefore

$$\nabla \chi = -\frac{\Phi_B}{2\pi s} \hat{\varphi} = -\frac{BR^2}{2r} \hat{\varphi} \quad (31)$$

Thus

$$\mathbf{A}' = \mathbf{A} + \nabla \chi = \begin{cases} \left(\frac{Br}{2} - \frac{BR^2}{2r} \right) \hat{\varphi} & \text{if } r \leq R \\ 0 & \text{if } r \geq R \end{cases} \quad (32)$$

Hence, under this gauge transformation, the vector potential $\mathbf{A}$ vanishes everywhere outside the cylinder.

$$\mathbf{A}'(\mathbf{r}) = 0 \quad \forall |\mathbf{r}| \geq R \quad (33)$$

(2.c)

$$\mathbf{B}' = \nabla \times \mathbf{A}' = \mathbf{B} - \frac{\Phi_B}{2\pi} \nabla \times \left(\frac{\hat{\varphi}}{r} \right). \quad (34)$$

For $r \neq 0$:

$$\nabla \times \left(\frac{\hat{\varphi}}{r} \right) = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{1}{r} \right) = 0. \quad (35)$$

What happens at $r = 0$?

We again consider a disk D_r parallel to the z -axis with radius r around the origin.

$$\int_{D_r} d^2\mathbf{S} \nabla \times \left(\frac{\hat{\varphi}}{r} \right) \stackrel{\text{Stokes}}{=} \oint_{\partial D_r} d\mathbf{l} \frac{\hat{\varphi}}{r} = \int_0^{2\pi} d\varphi = 2\pi. \quad (36)$$

Thus

$$\nabla \times \left(\frac{\hat{\varphi}}{r} \right) = \frac{\delta(r)}{r} \hat{\mathbf{z}} \quad (37)$$

$$\Rightarrow \mathbf{B}' = \mathbf{B} - \frac{\delta(r)}{r} \hat{\mathbf{z}}. \quad (38)$$

At first glance this seems confusing, since a gauge transformation should not change the magnetic field $\mathbf{B}$. However, one must note that our transformation $\mathbf{A} \rightarrow \mathbf{A} + \nabla\chi$ is only well-defined on $\mathbb{R}^3 \setminus \{\mathbf{r} | x = y = 0\}$. At the points $(0, 0, z)$ one cannot assign a unique value to χ , which leads to the magnetic field also being changed at these points. The gauge transformation is therefore not globally well-defined and cannot simply be applied to the entire space $\mathbb{R}^3$.

Therefore, the Aharonov–Bohm effect **cannot** be eliminated by a gauge transformation.

Problem 3 The Total Momentum of a Static Electromagnetic Field

When both an electric and a magnetic field are present, the momentum density of the electromagnetic field is given by

$$\mathcal{P} = \epsilon_0 \mathbf{E} \times \mathbf{B}. \quad (39)$$

Consider a spatially bounded distribution of time-independent charges and currents.

(3.a) Show that the total momentum can be written as

$$\mathbf{P} \equiv \int d^3\mathbf{r} \mathcal{P}(\mathbf{r}) = \epsilon_0 \mu_0 \int d^3\mathbf{r} \phi(\mathbf{r}) \mathbf{J}(\mathbf{r}) = \frac{1}{c^2} \int d^3\mathbf{r} \phi(\mathbf{r}) \mathbf{J}(\mathbf{r}). \quad (40)$$

(3.b) Give an example of a stationary, bounded charge and current distribution for which $\mathbf{P} \neq 0$.

Hint: It suffices to consider the simplest charge and current distributions side by side.

Solution:

(3.a)

$$\begin{aligned}\mathbf{P} &= \epsilon_0 \int d^3\mathbf{r} \mathbf{E} \times \mathbf{B} \\ &= -\epsilon_0 \int d^3\mathbf{r} \nabla\phi \times \mathbf{B}.\end{aligned}$$

We use the identity from the first problem set

$$\nabla \times (\phi \mathbf{u}) = \phi \nabla \times \mathbf{u} + \nabla\phi \times \mathbf{u}.$$

Therefore,

$$\mathbf{P} = \epsilon_0 \int d^3\mathbf{r} [\phi(\nabla \times \mathbf{B}) - \nabla \times (\phi \mathbf{B})].$$

Using Gauß' theorem on the second term, over a ball $B_R(0)$, gives

$$\begin{aligned}\int_{B_R(0)} d^3\mathbf{r} \nabla \times (\phi \mathbf{B}) &= \int_{\partial B_R(0)} d\mathbf{A} \times (\phi \mathbf{B}) \\ &= \int_{\partial B_R(0)} dA \phi \hat{\mathbf{r}} \times \mathbf{B}.\end{aligned}$$

Hence,

$$\begin{aligned}\mathbf{P} &= \epsilon_0 \int d^3\mathbf{r} \phi(\nabla \times \mathbf{B}) - \epsilon_0 \lim_{R \rightarrow \infty} \int_{S_R(0)} dA \phi \hat{\mathbf{r}} \times \mathbf{B} \\ &= \epsilon_0 \int d^3\mathbf{r} \phi(\nabla \times \mathbf{B}) \\ &= \epsilon_0 \mu_0 \int d^3\mathbf{r} \phi \mathbf{J}.\end{aligned}$$

The boundary term vanishes for spatially bounded charge and current distributions.

(3.b) Now consider a point charge q at the origin and a current loop of radius R at position $\mathbf{d} = D\hat{\mathbf{x}}$, with $D/R \gg 1$. Then we may approximate

$$\phi(\mathbf{r}) = \phi(\mathbf{d}) - (\mathbf{r} - \mathbf{d}) \cdot \mathbf{E}(\mathbf{d}).$$

Now note that

$$\begin{aligned}\nabla \cdot (r_\nu \mathbf{J}) &= r_\nu (\nabla \cdot \mathbf{J}) + \mathbf{J} \cdot \nabla r_\nu \\ &= J_\nu.\end{aligned}$$

It follows that

$$\begin{aligned}\int d^3\mathbf{r} J_\nu &= \int d^3\mathbf{r} \nabla \cdot (r_\nu \mathbf{J}) \\ &= \int d\mathbf{A} \cdot (r_\nu \mathbf{J}) \\ &= 0,\end{aligned}$$

since $\mathbf{J} \rightarrow 0$ at infinity.

Similarly, considering

$$\int d^3\mathbf{r} \nabla \cdot (r_\mu r_\nu \mathbf{J}) = 0,$$

we find

$$\int d^3\mathbf{r} r_\nu J_\mu = - \int d^3\mathbf{r} r_\mu J_\nu.$$

It follows that

$$\begin{aligned} \int d^3\mathbf{r} r_\ell J_\mu &= \frac{1}{2} \epsilon_{\ell\mu i} \int d^3\mathbf{r} (\mathbf{r} \times \mathbf{J})_i \\ &= \epsilon_{\ell\mu i} m_i, \end{aligned}$$

where

$$\mathbf{m} = \frac{1}{2} \int d^3\mathbf{r} \mathbf{r} \times \mathbf{J}$$

is the magnetic dipole moment.

Putting everything together,

$$\begin{aligned} \mathbf{P} &= \epsilon_0 \mu_0 \int d^3\mathbf{r} \phi \mathbf{J} \\ &\approx \epsilon_0 \mu_0 \int d^3\mathbf{r} [(\phi(\mathbf{d}) + \mathbf{d} \cdot \mathbf{E}(\mathbf{d})) \mathbf{J} - (\mathbf{r} \cdot \mathbf{E}(\mathbf{d})) \mathbf{J}] \\ &= -\epsilon_0 \mu_0 E_\ell(\mathbf{d}) \int d^3\mathbf{r} r_\ell J_\mu \hat{\mathbf{e}}_\mu \\ &= -\epsilon_0 \mu_0 E_\ell(\mathbf{d}) \epsilon_{\ell\mu i} m_i \hat{\mathbf{e}}_\mu \\ &= \epsilon_0 \mu_0 \mathbf{E}(\mathbf{d}) \times \mathbf{m}. \end{aligned}$$

Note that this solution is exact and non-vanishing for infinitesimal current loops with a magnetic dipole moment that is still finite.